

A claim about a population

Hypothesis testing uses sample evidence to assess a hypothesis: a claim about a population parameter.

Does a claim about average processing time concern μ or the observed sample mean?

The null hypothesis

The null hypothesis specifies the population claim assessed as the reference model.

For a claimed mean processing time of 12 minutes, how could we write H_0 ?

The alternative hypothesis

The alternative describes the population change or difference for which we seek evidence.

If the question is whether the mean exceeds 12 minutes, how should H_a be written?

Direction of the question

A left-tailed alternative uses $<$, a right-tailed alternative uses $>$, and a two-tailed alternative uses $\neq$.

Which alternative matches a question about any change from $\mu=12$?

Complementary hypotheses

For a claim that a proportion exceeds 0.4, OpenStax permits $H_0: p \leq 0.4$ with $H_a: p > 0.4$; calculations use the boundary $p = 0.4$.

How would you state hypotheses for a claim that a proportion is less than 0.4?

Mean or proportion hypotheses

A service advertises that 90% of requests finish on time.

Which parameter belongs in hypotheses about whether the advertised percentage is too high?

Choosing before looking

The research question should determine the alternative before inspecting the outcome.

What problem arises if we choose the tail only after seeing whether the sample mean is high or low?

The null sampling model

Under H_0 , a sampling model describes the statistic's variation if the null claim holds.

Why should this model concern sample means rather than individual processing times?

The Central Limit Theorem in testing

The Central Limit Theorem can justify an approximately normal sampling distribution for a mean when its conditions hold.

Does it say that individual observations must be normally distributed?

Choosing a reference distribution

Mean tests use a normal distribution with known population SD, or Student's t-distribution with estimated SD, under suitable assumptions.

Why does unknown population standard deviation affect the reference distribution?

A proportion test model

A proportion test starts with a binomial distribution. A normal approximation uses the null proportion and adequate expected success and failure counts.

Why should the null claim supply p when finding a p -value?

A test statistic

A test statistic summarizes how far sample evidence departs from the null model.

Why should distance from the null value be judged relative to standard error?

An assumption check

A simple random sample gives every sample of a fixed size the same chance. A valid test also needs its probability-model assumptions.

Could a very small p-value repair a strongly biased sample?

The p-value

A p-value is the null-model probability of results at least as extreme as observed, in the direction of the alternative.

Which assumption about H_0 is used when calculating it?

Rare-event reasoning

A rare result under a null model can count as evidence against that model, provided the sampling assumptions are reasonable.

Why does an unlikely result prompt scrutiny rather than prove the null hypothesis false?

One tail or two?

Synthetic result: the sample mean is above the null mean. The alternative is two-sided.

Would considering only equally high sample means capture all results at least as extreme?

Shading the p-value

A right-tailed test has an observed statistic to the right of the null center.

Which tail beyond the observed statistic represents its p-value?

A p-value is not a hypothesis probability

Synthetic result: a valid test reports $p\text{-value}=0.03$.

Does this assign a 3% probability to H_0 being true?

Significance level

The significance level, or level of significance, α sets the rejection rule and controls the Type I error rate under a valid test.

Why should α be chosen before examining the p-value?

A rejection decision

Synthetic valid test: $p\text{-value}=0.03$;
significance level $\alpha=0.05$.

What decision follows for H_0 ?

A nonrejection decision

Synthetic valid test: $p\text{-value}=0.18$;
significance level $\alpha=0.05$.

Does failing to reject H_0 establish that it is true?

The decision rule

For this course, follow OpenStax: reject when $p\text{-value} < \alpha$; otherwise do not reject. Apply the rule to unrounded values.

What decision follows when an exact p-value equals α ?

Type I error

A Type I error rejects a null hypothesis that is true.

If H_0 states that mean processing time is 12 minutes, what would a Type I error conclude?

Type II error

A Type II error fails to reject a null hypothesis that is false.

If the true mean exceeds 12 minutes, what would a Type II error miss?

Alpha and beta

Alpha (α) is the Type I error probability. Beta (β) is the Type II error probability for a specified false-null alternative.

If $\beta = 0.2$ at a particular alternative, what is the power?

Consequences of errors

H_0 says a printing process meets its quality target. Rejecting triggers an expensive shutdown. Failing to reject leaves it running.

How would you compare the consequences of Type I and Type II errors?

Power

Power of the test is $1 - \beta$: the probability of rejecting H_0 for a specified alternative that is true.

How might a larger sample help detect a small departure from the null value?

Statistical and practical importance

A small p-value measures evidence against a null model, rather than the size or importance of the difference.

Could a 0.02-minute difference be statistically significant but operationally unimportant?

An interval connection

For matching two-sided mean procedures, 95% confidence intervals excluding μ_0 correspond to rejecting $H_0: \mu = \mu_0$ at $\alpha = 0.05$.

Would a matching interval from 12.2 to 12.8 reject a null mean of 12?

A careful conclusion

Synthetic valid test rejects $H_0:\mu=12$ in favor of $H_a:\mu>12$.

How would you state the evidence without claiming proof or explaining why the mean changed?

Interpreting a supplied test result

Synthetic valid test: $H_0: p = 0.9$, $H_a: p < 0.9$, $p\text{-value} = 0.12$, $\alpha = 0.05$. Here p is the on-time fraction.

What decision and contextual conclusion follow without claiming that 90% has been proved?